

Shan Jiang

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[github](#) | [linkedin](#)

EDUCATION

M.S. Computational Science and Engineering (CSE)

Aug 2024 - Current

University of Texas at Austin

Austin, TX

Current coursework: High Performance Computing & Computer Architecture

B.S. Mathematics of Computation (Math&CS)

Awarded in Jun 2024

University of California, Los Angeles

Los Angeles, CA

GPA: 3.94/4.00, Cum Laude

Selected coursework: Data Structure, Algorithms, Java with Applications, Deep Learning, Linear Programming, Generative AI, Computer Vision, Machine learning, Optimization, Scientific Computing

SKILLS

Programming Languages C/C++ | Java | Python | C# | R | JavaScript | MATLAB | SQL | HTML | CSS | TypeScript

Tools/Technologies React | REST | WebRTC | ROS2 | Git | MPI | PyTorch | TensorFlow | Docker | Flask

EXPERIENCE

UT Austin Cockrell School of Engineering

Aug 2024 - Current

Graduate Teaching Assistant

Austin, TX

- Guided students in completing scientific computing projects focusing on the finite element method (**FEM**).
- Reviewed and provided feedback on students' **Python** programming, focusing on algorithm efficiency.

Howen Technologies

Jun 2024 - Aug 2024

Embedded Software Development Intern

Shenzhen, China

- Redesigned the Active Noise Suppression (ANS) module in **C++/C** using **WebRTC** by implementing a backward processing technique to extract clear audio in high-speed, extremely noisy environments.
- Integrated the enhanced ANS into the company's Mobile Digital Video Recorder (MDVR) products, resulting in a 60% improvement in audio clarity under challenging conditions.
- Optimized the parameters of the extended Kalman filter in **ROS2** robot_localization package and integrated it into MDVR products equipped with a six-axis sensor, enhancing the accuracy of vehicle tracking.

Polymath REU (Research Experiences for Undergraduates)

Jun 2023 - Aug 2023

Undergraduate Research Intern

Bethlehem, PA

- Developed a physics-based rod-chain model of train cars with derailment-resistant forces and a spring/dash-pot system to reduce node instability and determine stability thresholds under external forces.
- Simulated the model with an arbitrary number of rods using **MATLAB**.

New CarZone

Jun 2022 - Sep 2022

Software Engineer Intern

Nanjing, China

- Developed and tested over 10 **React** components with **TypeScript**, **HTML**, and **CSS** to improve the display and accessibility of automotive repair data, enhancing user interface quality within the digital store system.
- Customized **React Hooks** for seamless integration with **REST** APIs, enabling real-time updates of repair and maintenance records and improving overall data retrieval efficiency.

UCLA Program in Computing

Sep 2023 - Jun 2024

Course Learning Assistant

Los Angeles, CA

- Undertook a part-time role, grading **C++** coding assignments for the programming courses which focus on **Dynamic Data Structures** and **Object-Oriented** programming.

NOTABLE PROJECTS

VID-SYNC: A Simple YouTube Multi-Video Playback Syncing Tool

Austin, TX

- Built a **full-stack** web app using **Flask**, **HTML**, **CSS**, and **JavaScript** to automatically synchronize multiple videos through audio processing, achieving precise alignment of video streams for seamless playback.

Finite Element Solver for Poisson Equation

Austin, TX

- Developed a **C++** Poisson equation solver over 2D domain using Finite Element Method.
- Leveraged **CMake** for build automation and **Docker** for cross-platform consistency, with integration of **GDB**, **Valgrind**, **BOOST**, **Test**, and **GCOV** for comprehensive debugging, testing, and solution validation.